

Is Lattice Reduction Necessary for Vector Perturbation Precoding?

Dominik Semmler, Wolfgang Utschick, and Michael Joham

School of Computation, Information and Technology, Technical University of Munich, 80333 Munich, Germany

email: {dominik.semmler,utschick,joham}@tum.de

Abstract—Vector perturbation (VP) precoding is an effective nonlinear precoding technique in the downlink (DL) with modulo channels. Especially, when combined with Lattice reduction (LR), low-complexity algorithms achieve very promising performances, outperforming other popular nonlinear precoding techniques like Tomlinson-Harashima precoding (THP). However, these results are based on the uncoded symbol error rate (SER) or uncoded bit error rate (BER). We show that when using the mutual information as the figure of merit, the observation is fundamentally different and that these algorithms generally do not outperform THP. Within the expression of the mutual information, a rate allocation matrix can be incorporated, which has not received much attention so far. In this article, we derive the optimal choice of this matrix for different algorithms, and we show that this matrix is indeed crucial for the performance, especially for ill-conditioned channels. Furthermore, when using an optimized choice of this matrix, we show that the classical LR-aided algorithms cannot exceed the rate of THP, highlighting the effectiveness of the THP method. This concept can be generalized to a whole class of algorithms for which LR yields no improvement. We derive the corresponding properties and categorize various algorithms accordingly.

Index Terms—THP, Lattice Reduction, Vector Perturbation, Rate

I. INTRODUCTION

Dirty paper coding (DPC) is known to achieve the capacity of the multiple-input multiple-output (MIMO) broadcast channel (BC) [1]. However, as its implementation complexity is considered to be prohibitive, approximations like linear precoding are often adopted. In [2], VP has been proposed to improve upon linear precoding by introducing an integer vector at the base station (BS) which reduces the energy of the transmit symbol. This vector can be optimized without introducing interference because the receivers are equipped with modulo operators, which remove any perturbations by integers. Optimizing this integer vector is a lattice closest vector problem, and the sphere decoding (SD) algorithm (see, e.g., [3], [4], [5], [6], [7]) gives the optimal solution. While this has been discussed in [2], the SD algorithm has an exponential complexity, rendering it impractical. Hence, polynomial-time algorithms are preferred, and in [8], various algorithms, like the nearest plane (NP) and the rounding off (RO) algorithms, have been proposed. Most importantly, the effectiveness of applying LR prior to the application of the NP or RO method has been demonstrated in [8]. LR (see [9]) is an elegant way for optimizing a lattice problem, transforming the original lattice basis into one with short and nearly-orthogonal basis vectors without changing the underlying problem. After this reduction, already very simple methods are significantly improved and achieve a good performance [8]. The popular THP method (see [10], [11], [12], [13], [14]) can also be interpreted as a particular sub-optimal technique for VP, resulting in a specific integer vector. This optimization concept of THP is very similar to the NP method [8], and it can also be combined with LR.

It has already been shown (see, e.g., [15], [16], [17]) that LR techniques clearly improve upon conventional methods, including THP, albeit allowing a polynomial complexity. However, these results

were based on the uncoded error rates, whereas in this article, we consider the mutual information representing the maximum number of bits that can be transferred per channel use by the communication system for an arbitrarily small error probability. In [18], VP has already been analyzed when considering the mutual information as the metric. [18] is primarily based on the zero-forcing (ZF) precoder whereas the analysis was extended to minimum mean square error (MMSE) approaches in [19]. Furthermore, in [20], VP was generalized to a processing over multiple time instances where also the mutual information with ZF precoders was analyzed. We are also focusing on ZF precoders where the precoding matrix is given by the pseudoinverse of the channel. However, there is an important design choice that we will show leading to fundamentally different results. Instead of just using the plain pseudoinverse as the ZF precoder, a diagonal matrix can be incorporated in front of the pseudoinverse without introducing any interference. In [18], this has already been discussed by using a lower bound on the symbol energy (see [21]). When considering the high-signal to noise ratio (SNR) regime, where all users are allocated, the mutual information results in an expression independent of the rate allocation matrix when using the bound in [21] (see [18]). Hence, it follows that the diagonal allocation matrix has no impact in this regime. Nevertheless, this observation is based on a bound, and in this article, we show that when considering actual algorithms (including the optimal solution found by SD), the choice of this diagonal allocation matrix is fundamental. Hence, also in the high-SNR regime, it has a strong impact whether the diagonal allocation matrix in front of the pseudoinverse is optimized or not.

Throughout the article, the mutual information will be our metric and, hence, the diagonal allocation matrix is optimized w.r.t. the mutual information. When using the optimized allocation matrix, all algorithms lead to a significantly improved performance. This also holds for the SD algorithm in our scenarios. The differences are especially pronounced when analyzing ill-conditioned channels. Beyond the improved performance, the optimized allocation matrix results in the interesting aspect that the arising lattice problem has a very special structure. In particular, the specific choice of the allocation matrix leads to a lattice problem where the basis matrix of the lattice is unitriangular. We will show that when having such a unitriangular basis matrix, LR methods have to be re-evaluated. The popular LR-aided NP and RO techniques [8] are not as powerful anymore and, surprisingly, cannot outperform the simple NP algorithm or THP. This aligns w.r.t. the NP algorithm with [22], where the effectiveness of the Lenstra–Lenstra–Lovász (LLL) reduction has been analyzed in a detection scenario with Gaussian noise. It has been demonstrated that, in general, the success probability of the NP algorithm increases when the LLL reduction is applied. Only under specific circumstances, it has been shown that the LLL reduction does not lead to an increase in the success probability. The unitriangular basis matrix, which results from our optimization of the mutual information for VP, falls under such a circumstance and, hence, the

success probability of the NP algorithm is not expected to improve in case of a unitriangular basis matrix. However, the analysis in [22] for the NP algorithm is based on the success probability in a detection scenario. The success probability is not available in our case as we don't deal with a detection problem. Instead, we consider the symbol energy minimization when applying VP in the DL. Nevertheless, we will show in this article that the NP algorithm will not improve when LR is applied in the case of the unitriangular basis matrix.

These observations can be generalized to more advanced algorithms. In particular, a class of algorithms (greedy-based schemes) can be defined that cannot be improved with LR in case the lattice basis matrix is unitriangular. While a number of algorithms can be considered for this characterization, we focus on two very powerful and popular algorithms beyond the SD algorithm. These are the K-Best (see, e.g., [23], [24], [25], [26], [27], [28], [29]) and Fixed-Complexity Sphere Decoder (FSD) (see, e.g., [30], [31], [32], [33], [34], [35], [36]) algorithms which provide a better performance than the simple NP and RO methods.

In summary, this paper has the following contributions:

- We derive the optimal rate allocation matrix for various algorithms, most importantly for THP and the NP algorithm. Additionally, we show mathematically as well as numerically that this matrix has a fundamental impact on the performance. In contrast to the uncoded SER or BER, when considering the mutual information, we demonstrate the importance of an optimized allocation matrix. We extend the observations to more complex algorithms like the SD algorithm and show that, also in this case, the rate allocation matrix is essential. This is especially pronounced for ill-conditioned channels. If this allocation matrix is omitted and the standard version of SD is used as it is proposed in the literature, we show that conventional THP with the appropriate allocation matrix actually outperforms the SD algorithm due to the choice of the allocation matrix.
- Furthermore, we show that when the optimized rate allocation matrix is employed for THP or the NP algorithm, LR cannot improve the performance. In the existing literature, it is assumed that LR drastically improves the performance and even simple algorithms like the RO or NP combined with LLL reduction clearly outperform THP. However, this is only true for the uncoded SER and BER metrics. When considering the mutual information, we prove that when combining the NP or the RO algorithm with LLL reduction, the mutual information cannot exceed the one of simple THP (or NP without LR). The LLL-aided NP is also shown to achieve the exact same solution as the conventional NP algorithm.
- There are more algorithms beyond the NP and the RO algorithm for which LLL reduction has no impact. We give a generalization of this effect and analyze which properties the algorithms have to fulfill such that LLL reduction can be neglected. Additionally, we categorize various algorithms according to these properties and evaluate their performance with and without LR. We also show that the LLL-aided K-Best method leads to the exact same solution as the conventional K-Best algorithm.

II. SYSTEM MODEL

We consider K single-antenna users being served by one BS having N antennas. The channel of the k -th user is denoted as $\mathbf{h}_{\mathcal{C},k}^H \in \mathbb{C}^{1 \times N}$, $y_{\mathcal{C},k}$ is the complex received signal of user k , and

$\mathbf{x}_{\mathcal{C}}$ is the complex transmit symbol at the base station. This results in the system model

$$\mathbf{y}_{\mathcal{C}} = \mathbf{H}_{\mathcal{C}} \mathbf{x}_{\mathcal{C}} + \mathbf{n}_{\mathcal{C}} \in \mathbb{C}^K \quad (1)$$

where $\mathbf{n}_{\mathcal{C}} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{I})$ is the complex Gaussian noise vector. By stacking the real and imaginary parts as $\mathbf{y} = [\text{Re}(\mathbf{y}_{\mathcal{C}}^T), \text{Im}(\mathbf{y}_{\mathcal{C}}^T)]^T$, $\mathbf{x} = [\text{Re}(\mathbf{x}_{\mathcal{C}}^T), \text{Im}(\mathbf{x}_{\mathcal{C}}^T)]^T$, and $\mathbf{n} = [\text{Re}(\mathbf{n}_{\mathcal{C}}^T), \text{Im}(\mathbf{n}_{\mathcal{C}}^T)]^T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{\frac{1}{2}})$, we arrive at the equivalent real-valued representation

$$\mathbf{y} = \mathbf{H} \mathbf{x} + \mathbf{n} \in \mathbb{R}^{2K}, \quad \text{with } \mathbf{H} = \begin{bmatrix} \text{Re}(\mathbf{H}_{\mathcal{C}}) & -\text{Im}(\mathbf{H}_{\mathcal{C}}) \\ \text{Im}(\mathbf{H}_{\mathcal{C}}) & \text{Re}(\mathbf{H}_{\mathcal{C}}) \end{bmatrix}. \quad (2)$$

Considering a modulo channel, each receiver is equipped with a modulo operator located after the receive filter. Additionally, the transmit symbol is constructed as

$$\mathbf{x} = \rho \bar{\mathbf{x}} = \rho \mathbf{P}(\mathbf{s} + \mathbf{a}) \in \mathbb{R}^{2N} \quad (3)$$

where $\mathbf{P} \in \mathbb{R}^{2N \times 2K}$ is the precoder, $\mathbf{s} \in \mathbb{R}^{2K}$ is the symbol vector where s_k and s_{k+K} is respectively the real and imaginary part of the symbol for user k . The characteristic of the modulo channel is the integer vector $\mathbf{a} \in \mathbb{Z}^{2K}$, which is added to the symbol vector and is designed to optimize the performance. Furthermore, the scaling $\rho = \sqrt{P_{\text{Tx}}/\mathbb{E}[\|\bar{\mathbf{x}}\|^2]}$ is introduced such that the transmit power constraint $\mathbb{E}[\|\mathbf{x}\|^2] = \mathbb{E}[\|\bar{\mathbf{x}}\|^2]\rho^2 \leq P_{\text{Tx}}$ is fulfilled with equality. Combining these definitions, the system model reads as

$$\tilde{\mathbf{s}} = \text{Mod}(\mathbf{G} \mathbf{H} \mathbf{P}(\mathbf{s} + \mathbf{a})\rho + \mathbf{G} \mathbf{n}) \quad (4)$$

where $\mathbf{G}$ is the diagonal receive filter and $\tilde{\mathbf{s}}$ is the vector after the modulo operators. As a precoder, we choose the ZF solution

$$\mathbf{P} = \mathbf{H}^+ \mathbf{D} \quad (5)$$

where $\mathbf{H}^+$ is the pseudoinverse of the channel and $\mathbf{D} = \text{diag}(d_1, \dots, d_{2K})$ is the diagonal rate allocation matrix which we show to be an essential aspect for the performance. Considering a general matrix $\mathbf{D}$, the receive filter is given by

$$\mathbf{G} = \mathbf{D}^{-1} \frac{1}{\rho} \quad (6)$$

after which the received signal can be expressed as

$$\tilde{\mathbf{s}} = \text{Mod}(\mathbf{s} + \mathbf{D}^{-1} \mathbf{n} / \rho) \quad (7)$$

where the integer vector $\mathbf{a}$ vanishes due to the modulo operations. It has been shown in [37] that the optimal distribution of s_i is a uniform distribution between -0.5 and 0.5 denoted as $s_i \sim \mathcal{U}(-0.5, 0.5)$. Following [18], the mutual information reads as

$$I(\tilde{\mathbf{s}}; \mathbf{s}) = - \sum_{k=1}^{2K} h\left(\text{Mod}(s_k + n_k/(\rho d_k))\right). \quad (8)$$

In this article, our main focus is on the high-SNR regime where the asymptote of $I(\tilde{\mathbf{s}}; \mathbf{s})$ (see [18, Section III]) can be expressed as

$$R^{\text{HSNR}} = K \log_2 \left(\frac{P_{\text{Tx}}}{\pi e \mathbb{E}[\|\bar{\mathbf{x}}\|^2]} \sqrt{\prod_{k=1}^{2K} d_k^2} \right). \quad (9)$$

So far, we have been following [18], [38] for the system model. This high-SNR asymptote (9) is the utility of this article and jointly optimizing this expression w.r.t. the scaling $\mathbf{D}$ and the integer values $\mathbf{a}$ is analyzed in the following. When optimizing this expression w.r.t. $\mathbf{D}$, it is important to note that $\mathbf{D}$ also appears inside $\mathbb{E}[\|\bar{\mathbf{x}}\|^2]$ which makes the optimization non-trivial. This joint optimization is typically omitted in the literature, and $\mathbf{D}$ is chosen to have fixed

values, particularly, $\mathbf{D} = \mathbf{I}$ (see, e.g., [2]). [18] addressed the joint optimization of $\mathbf{D}$ and $\mathbf{a}$ by using the bound of [21, Lemma 1]

$$\mathbb{E}[\|\bar{\mathbf{x}}\|^2] \geq \frac{K\Gamma(K+1)^{\frac{1}{K}}}{(K+1)\pi} \det(\mathbf{D}(\mathbf{H}\mathbf{H}^T)^{-1}\mathbf{D})^{\frac{1}{2K}}. \quad (10)$$

Substituting this expression into (9) results in the upper bound

$$R_{\text{UB}}^{\text{HSNR}} = \log_2 \det(\mathbf{H}_C \mathbf{H}_C^H \bar{\mathbf{p}}) - K \log_2 \left(\frac{e\Gamma(K+1)^{\frac{1}{K}}}{K+1} \right) \quad (11)$$

with $\bar{p} = P_{\text{Tx}}/K$. Hence, following [18], it can be observed that optimizing the scaling matrix $\mathbf{D}$ is not important at high-SNR as (11) does not depend on $\mathbf{D}$. However, the importance of this result highly depends on the tightness of the bound in (10). We will show in the following that when using actual algorithms, including the optimal SD algorithm, the choice of $\mathbf{D}$ does matter.

III. CONVENTIONAL ALGORITHMS

A variety of different algorithms can be used to minimize

$$\mathbb{E}[\|\bar{\mathbf{x}}\|^2] = \mathbb{E}[\|\mathbf{H}^+ \mathbf{D}(\mathbf{s} + \mathbf{a})\|^2] \quad (12)$$

w.r.t. the integer values $\mathbf{a}$. The optimal choice can be found by SD and has been analyzed in [2] with the identity scaling $\mathbf{D} = \mathbf{I}$. Due to its exponential complexity, we focus on the algorithms in [8], where at first we consider the conventional methods without LR.

A. Rounding Off

The RO procedure (see [8]) is probably the simplest method. Here, each a_i is chosen as the nearest integer, i.e.,

$$a_i^{\text{RO}} = \lceil s_i \rceil = 0 \quad (13)$$

where $\lceil \bullet \rceil$ is the rounding operation to the nearest integer. Therefore, the solution results in the zero vector as the symbols s_i are assumed to be distributed between -0.5 and 0.5 . As $\mathbf{a} = \mathbf{0}$, the solution results in classic linear zero-forcing beamforming.

B. Nearest Plane

The NP algorithm of [8] is a considerable improvement. With the QR decomposition $\mathbf{H}^+ \mathbf{D} = \mathbf{Q}\mathbf{R}\mathbf{D} = \mathbf{Q}\tilde{\mathbf{R}}$, the integers a_i are now successively chosen as

$$a_i^{\text{NP}} = - \left\lceil s_i + \sum_{j>i} \frac{\tilde{r}_{ij}}{\tilde{r}_{ii}} (s_j + a_j) \right\rceil \quad (14)$$

by starting with $i = 2K$ until $i = 1$.

C. THP

Finally, also the popular THP algorithm can be interpreted as a method which optimizes (12) w.r.t. the integer vector $\mathbf{a}$. Interestingly, it follows the same procedure as the NP algorithm. The only difference is that THP works on a lower triangular matrix, whereas the NP algorithm uses an upper triangular matrix. Hence, for THP, we use an LQ decomposition $\mathbf{H} = \mathbf{L}\mathbf{Q}$ and with $\mathbf{H}^+ \mathbf{D} = \mathbf{Q}^T \mathbf{L}^{-1} \mathbf{D} = \tilde{\mathbf{L}}^{-1}$, the integers a_i are successively updated as

$$a_i^{\text{THP}} = - \left\lceil s_i + \sum_{j<i} \frac{\tilde{l}_{ij}^{-1}}{\tilde{l}_{ii}^{-1}} (s_j + a_j) \right\rceil \quad (15)$$

by starting with $i = 1$ until $i = 2K$ where $\tilde{l}_{ij}^{-1}$ is the ij -th entry of $\tilde{\mathbf{L}}^{-1}$.

D. Overview

It is important to note that we generalized the algorithms above w.r.t. an arbitrary scaling matrix $\mathbf{D}$. Very often, the algorithms are given with the identity scaling $\mathbf{D} = \mathbf{I}$. Besides, the THP algorithm is also given with the scaling $\mathbf{D} = \text{diag}(\mathbf{l})$ where $\mathbf{l}$ is the diagonal of $\mathbf{L}$ (see, e.g., [39] for both scalings). We will use the generalized versions with an arbitrary $\mathbf{D}$, and we will derive the optimal solution of $\mathbf{D}$ for each of the algorithms in the following. As the THP and the NP algorithms follow the same concept and have the same performance, we give all proofs only once for the NP algorithm. However, all results can be immediately extended to THP.

IV. PERFORMANCE OF CONVENTIONAL ALGORITHMS

In this section, we derive the optimal scaling matrix $\mathbf{D}$ and the corresponding rates for the different algorithms in closed form. For the derivations, we need the following lemma.

Lemma 1. *For two independent random variables X and N where $X \sim \mathcal{U}(-0.5, 0.5)$, the distribution of $Y = \text{Mod}(X + N)$ is again uniform between -0.5 and 0.5 .*

Proof. This is a well-known result. We gave a short proof in [40]. $\square$

A. Rounding Off

The RO algorithm coincides with the linear zero-forcing beamforming solution. Even though we are assuming a uniform distribution instead of a Gaussian distribution, this does not affect the optimal scaling $\mathbf{D}$. Also, the rate resembles that of linear zero-forcing at high-SNR with the difference that we have a shaping loss of $K \log_2 \left(\frac{\pi e}{6} \right)$ bits as the uniform distribution of the data symbols is not optimal. This observation is summarized in the following proposition.

Proposition 1. *The optimal scaling matrix $\mathbf{D}$ when using the conventional RO algorithm is given by*

$$\mathbf{D} = \text{diag}((\mathbf{H}\mathbf{H}^T)^{-1})^{\frac{1}{2}}. \quad (16)$$

For this optimal choice of the scaling matrix, the RO rate reads as

$$R_{\text{RO}}^{\text{HSNR}} = \sum_{i=1}^K \log_2 \left(\frac{\bar{p}}{e_i^T (\mathbf{H}_C \mathbf{H}_C^H)^{-1} e_i} \right) - K \log_2 \left(\frac{\pi e}{6} \right). \quad (17)$$

Proof. For the RO solution ($\mathbf{a} = \mathbf{0}$), the transmit signal reduces to

$$\mathbf{x} = \mathbf{H}^+ \mathbf{D}(\mathbf{s} + \mathbf{a}) = \mathbf{H}^+ \mathbf{D} \mathbf{s}. \quad (18)$$

It follows that the signal power is given by

$$\mathbb{E}[\|\mathbf{x}\|^2] = \text{tr}(\mathbf{H}^+ \mathbf{D} \mathbb{E}[\mathbf{s} \mathbf{s}^T] \mathbf{D} \mathbf{H}^{+,T}) = \text{tr}(\mathbf{D}(\mathbf{H}\mathbf{H}^T)^{-1} \mathbf{D})/12 \quad (19)$$

as the symbols s_i are independent of each other and have variance $\mathbb{E}[s_i^2] = \frac{1}{12}$. Using the arithmetic-geometric mean inequality yields

$$\begin{aligned} \text{tr}(\mathbf{D}(\mathbf{H}\mathbf{H}^T)^{-1} \mathbf{D}) &= \sum_{i=1}^{2K} d_i^2 e_i^T (\mathbf{H}\mathbf{H}^T)^{-1} e_i \\ &\geq 2K \sqrt[2K]{\prod_{i=1}^{2K} d_i^2 e_i^T (\mathbf{H}\mathbf{H}^T)^{-1} e_i} \end{aligned} \quad (20)$$

we can upper bound the rate [cf. (9)] as

$$R_{\text{RO}}^{\text{HSNR}} \leq K \log_2 \left(\frac{12P_{\text{Tx}}}{2K\pi e} \sqrt[2K]{\prod_{i=1}^{2K} (e_i^T (\mathbf{H}\mathbf{H}^T)^{-1} e_i)^{-1}} \right) \quad (21)$$

which can be rewritten as in (17). This upper bound is achieved for $d_i = \pm 1/\sqrt{e_i^T (\mathbf{H}\mathbf{H}^T)^{-1} e_i}$. Hence, the scaling in (16) is an optimal solution. Note that the sign of d_i can be arbitrarily chosen and does not affect the rate. $\square$

B. Nearest Plane

The NP algorithm improves on the simple RO procedure. Interestingly, in comparison to RO, which has the same performance as linear precoding with an additional shaping loss, the NP algorithm has the same performance as DPC with the same shaping loss. This is shown in the following, where we also give the optimal matrix $\mathbf{D}$.

Proposition 2. *The optimal scaling matrix $\mathbf{D}$, when using the conventional NP Algorithm, is given by*

$$\mathbf{D} = \text{diag}^{-1}(\mathbf{r}) \quad (22)$$

where $\mathbf{r}$ is the diagonal of $\mathbf{R}$ from the QR decomposition $\mathbf{H}^+ = \mathbf{Q}\mathbf{R}$. For this optimal choice of the scaling matrix, the rate reads as

$$R_{\text{NP}}^{\text{HSNR}} = \log_2 \det(\bar{p}\mathbf{H}_c\mathbf{H}_c^H) - K \log_2\left(\frac{\pi e}{6}\right). \quad (23)$$

Proof. With the QR decomposition $\mathbf{H}^+ = \mathbf{Q}\mathbf{R}$, we arrive at

$$\mathbf{x} = \mathbf{H}^+ \mathbf{D}(\mathbf{s} + \mathbf{a}) = \mathbf{Q}\mathbf{R}\mathbf{D}(\mathbf{s} + \mathbf{a}). \quad (24)$$

According to the NP algorithm in (14), we find

$$\begin{aligned} \mathbf{e}_i^T \mathbf{R}\mathbf{D}(\mathbf{s} + \mathbf{a}) &= r_{ii}d_i \left(\sum_{j>i} \frac{r_{ij}d_j}{r_{ii}d_i} (s_j + a_j) + s_i + a_i \right) \\ &= r_{ii}d_i \text{Mod}(w_i + s_i) \end{aligned} \quad (25)$$

where the last line follows from (14) with $w_i = \sum_{j>i} \frac{r_{ij}d_j}{r_{ii}d_i} (s_j + a_j)$. According to Lemma 1, as s_i is independent of w_i , also $\text{Mod}(w_i + s_i) \sim \mathcal{U}(-0.5, 0.5)$. It follows that $\mathbb{E}[|\mathbf{e}_i^T \mathbf{R}\mathbf{D}(\mathbf{s} + \mathbf{a})|^2] = r_{ii}^2 d_i^2 \frac{1}{12}$ and, therefore, we get for $\mathbb{E}[\|\bar{\mathbf{x}}\|^2]$

$$\mathbb{E}[\|\mathbf{Q}\mathbf{R}\mathbf{D}(\mathbf{s} + \mathbf{a})\|^2] = \mathbb{E}[\|\mathbf{R}\mathbf{D}(\mathbf{s} + \mathbf{a})\|^2] = \frac{1}{12} \sum_{i=1}^{2K} r_{ii}^2 d_i^2. \quad (26)$$

Applying the arithmetic-geometric mean inequality

$$\sum_{i=1}^{2K} r_{ii}^2 d_i^2 \geq 2K \sqrt[2K]{\prod_{i=1}^{2K} r_{ii}^2 d_i^2} \quad (27)$$

we can upper bound the rate as [cf. (9)]

$$R_{\text{NP}}^{\text{HSNR}} \leq K \log_2 \left(\frac{12P_{\text{Tx}}}{2K\pi e} \sqrt[2K]{\det(\mathbf{H}\mathbf{H}^T)} \right) \quad (28)$$

which can be rewritten as in (23). This upper bound is achieved for $d_i = \pm 1/r_{ii}$ and, hence, the scaling in (22) is optimal. $\square$

Corollary 1. *The same Rate can be achieved by conventional THP,*

$$R_{\text{THP}}^{\text{HSNR}} = R_{\text{NP}}^{\text{HSNR}}. \quad (29)$$

THP follows the same successive procedure as the NP algorithm, with the only difference that a lower instead of an upper triangular matrix is used. For THP, the optimal scaling matrix is, therefore,

$$\mathbf{D} = \text{diag}(\mathbf{l}) \quad (30)$$

where $\mathbf{l}$ is the diagonal of the LQ decomposition $\mathbf{H} = \mathbf{L}\mathbf{Q}$ (which means that $\mathbf{H}^+ = \mathbf{Q}^T \mathbf{L}^{-1}$).

V. PERFORMANCE OF LLL-AIDED ALGORITHMS

It has to be noted that the RO and the NP algorithms are typically combined with LR reduction (see [8]) by using the LLL algorithm [9]. This clearly improves the performance of the algorithms and has been demonstrated w.r.t the SER and BER in [15], [16], [17]. In this case, the solution of the RO method is not the zero-vector anymore and, hence, the LLL-aided RO improves upon the linear zero-forcing method. However, while the RO procedure is improved, we will see that the solution of the NP (or THP) cannot be improved by simply using LR. To this end, we need the following lemma.

Lemma 2. *Let $\mathbf{B}$ be the basis of a lattice $\mathcal{L}$ with QR decomposition $\mathbf{B} = \mathbf{Q}\mathbf{R}$ and let $\tilde{\mathbf{B}} = \mathbf{B}\mathbf{T} = \tilde{\mathbf{Q}}\tilde{\mathbf{R}}$ be the QR decomposition of the LLL-reduced basis $\tilde{\mathbf{B}}$, where $\mathbf{T}$ is a unimodular integer matrix. Then, if $|r_{ii}| \leq |r_{jj}|$, for $i < j$, it follows that $\tilde{r}_{ii} = r_{ii} \forall i$ and that $\mathbf{T}$ is an upper triangular matrix with $t_{ii} = 1, \forall i$.*

Proof. We follow the LLL algorithm of [9] with the code given in [41] where k is iterated from $k = 2$ until $k = n$. The algorithm is initialized with $k = 2$, $\mathbf{T} = \mathbf{I}$, $\tilde{\mathbf{Q}} = \mathbf{Q}$, and $\tilde{\mathbf{R}} = \mathbf{R}$. According to the LLL-algorithm, the k -th columns of $\tilde{\mathbf{R}}$ and $\mathbf{T}$ are updated as

$$\tilde{\mathbf{r}}_{1:l,k} = \tilde{\mathbf{r}}_{1:l,k} - \left\lfloor \frac{\tilde{r}_{lk}}{\tilde{r}_{ll}} \right\rfloor \tilde{\mathbf{r}}_{1:l,l} \quad \text{and} \quad \mathbf{t}_k = \mathbf{t}_k - \left\lfloor \frac{\tilde{r}_{lk}}{\tilde{r}_{ll}} \right\rfloor \mathbf{t}_l \quad (31)$$

where the variable l is decreased from $k - 1$ to 1 (see [41]). Combining (31) with the fact that $l \leq k - 1$, we can see that only the first $k - 1$ entries of $\tilde{\mathbf{r}}_k$ and $\mathbf{t}_k$ are modified. Hence, the upper triangular structure of $\tilde{\mathbf{R}}$ and $\mathbf{T}$ is preserved with this step. Additionally,

$$\tilde{r}_{k-1,k-1}^2 \leq \tilde{r}_{kk}^2 + \tilde{r}_{k-1,k}^2 \quad (32)$$

holds as $\tilde{r}_{k-1,k-1}^2 \leq \tilde{r}_{kk}^2$ by assumption. Therefore, the current iteration is completed [41] and k is incremented to $k + 1$. In addition to the upper triangular structure, the diagonal of $\mathbf{T}$ and $\tilde{\mathbf{R}}$ was preserved. These conservations hold for all iterations and, hence, the final matrices $\tilde{\mathbf{R}}$ and $\mathbf{T}$ are upper triangular with $t_{ii} = 1$, $\tilde{r}_{ii} = r_{ii}, \forall i$. $\square$

Remark 1. *Please note that if the scaling in (22) is used, the matrix $\mathbf{R} \text{diag}^{-1}(\mathbf{r})$ has only ones on the diagonal and, hence, the condition in Lemma 2 is fulfilled.*

A. LLL-aided Nearest Plane

We focus on the NP algorithm where, intuitively, LR should give an improvement to the conventional algorithm. However, when using the NP method with the optimized scaling (22), we can show with Lemma 2 that combining it with LLL reduction does not lead to a higher rate. This is summarized in the following Theorem.

Theorem 1. *LLL-aided NP leads to exactly the same solution as conventional NP when considering the optimized scaling in (22), i.e.,*

$$\mathbf{a}^{\text{LLL-NP}} = \mathbf{a}^{\text{NP}}, \quad \text{and} \quad R_{\text{NP-LLL}}^{\text{HSNR}} = R_{\text{NP}}^{\text{HSNR}}. \quad (33)$$

Proof. We apply the LLL reduction with the unimodular integer matrix $\mathbf{T}$ (which in this case is upper-unitriangular, see Lemma 2)

$$\mathbf{x} = \mathbf{H}^+ \mathbf{D} \mathbf{T} (\mathbf{T}^{-1} \mathbf{s} + \mathbf{T}^{-1} \mathbf{a}) = \tilde{\mathbf{Q}} \tilde{\mathbf{R}} (\tilde{\mathbf{s}} + \tilde{\mathbf{a}}) \quad (34)$$

where $\tilde{\mathbf{s}} = \mathbf{T}^{-1} \mathbf{s}$, $\tilde{\mathbf{a}} = \mathbf{T}^{-1} \mathbf{a}$, and $\tilde{\mathbf{Q}} \tilde{\mathbf{R}}$ being the QR decomposition of $\mathbf{H}^+ \mathbf{D} \mathbf{T}$. According to (14) of the NP algorithm, the integers $\tilde{a}_i$ are successively chosen as $\tilde{a}_i = - \left\lceil \tilde{s}_i + \sum_{j>i} \tilde{r}_{ij} (\tilde{s}_j + \tilde{a}_j) \right\rceil$. We assume now that all $\tilde{a}_j$ with $j > i$ have been obtained, and now $\tilde{a}_i$ is optimized. To this end, the matrices $\mathbf{T}$ and $\mathbf{R}$ are written as $\mathbf{T} = \begin{bmatrix} T_{<i,a} & T_{<i,b} \\ 0 & T_i \end{bmatrix}$ and $\mathbf{R} = \begin{bmatrix} R_{<i,a} & R_{<i,b} \\ 0 & R_i \end{bmatrix}$ where T_i and R_i

are the upper-triangular blocks starting from the i -th row. Due to the triangular structure, we have $\mathbf{T}^{-1} = \begin{bmatrix} \mathbf{T}_{<i,a}^{-1} & -\mathbf{T}_{<i,a}^{-1}\mathbf{T}_{<i,b}\mathbf{T}_i^{-1} \\ \mathbf{0} & \mathbf{T}_i^{-1} \end{bmatrix}$ and $\mathbf{RT} = \begin{bmatrix} \mathbf{R}_{<i,a}\mathbf{T}_{<i,a} & \mathbf{R}_{<i,a}\mathbf{T}_{<i,b} + \mathbf{R}_{<i,b}\mathbf{T}_i \\ \mathbf{0} & \mathbf{R}_i\mathbf{T}_i \end{bmatrix}$ with the submatrices

$$\mathbf{T}_i = \begin{bmatrix} 1 & \mathbf{t}_i^T \\ \mathbf{0} & \mathbf{T}_{i+1} \end{bmatrix}, \quad \mathbf{T}_i^{-1} = \begin{bmatrix} 1 & -\mathbf{t}_i^T\mathbf{T}_{i+1}^{-1} \\ \mathbf{0} & \mathbf{T}_{i+1}^{-1} \end{bmatrix}, \quad (35)$$

$$\mathbf{R}_i = \begin{bmatrix} 1 & \mathbf{r}_i^T \\ \mathbf{0} & \mathbf{R}_{i+1} \end{bmatrix}, \quad \text{and} \quad \mathbf{R}_i\mathbf{T}_i = \begin{bmatrix} 1 & \mathbf{t}_i^T + \mathbf{r}_i^T\mathbf{T}_{i+1} \\ \mathbf{0} & \mathbf{R}_{i+1}\mathbf{T}_{i+1} \end{bmatrix}. \quad (36)$$

As $\mathbf{T}$, $\mathbf{T}^{-1}$, and $\mathbf{R}$ are upper triangular matrices, only the blocks $\mathbf{T}_i$ and $\mathbf{R}_i$ have to be considered for optimizing $\tilde{\mathbf{a}}_i$ and we arrive at

$$|\mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2 = |\tilde{\mathbf{a}}_i + \tilde{\mathbf{s}}_i + (\mathbf{t}_i^T + \mathbf{r}_i^T\mathbf{T}_{i+1})(\tilde{\mathbf{s}}_{>i} + \tilde{\mathbf{a}}_{>i})|^2. \quad (37)$$

Minimizing this expression w.r.t. $\tilde{\mathbf{a}}_i$ results in

$$\tilde{\mathbf{a}}_i = -\left[\tilde{\mathbf{s}}_i + (\mathbf{t}_i^T + \mathbf{r}_i^T\mathbf{T}_{i+1})(\tilde{\mathbf{s}}_{>i} + \tilde{\mathbf{a}}_{>i})\right]. \quad (38)$$

With $\mathbf{a} = \mathbf{T}\tilde{\mathbf{a}}$ and exploiting the matrix structures in (35) and (36), we have

$$\begin{aligned} \mathbf{a}_i &= \mathbf{t}_i^T\mathbf{T}_{i+1}^{-1}\mathbf{a}_{>i} - \left[\tilde{\mathbf{s}}_i + (\mathbf{t}_i^T + \mathbf{r}_i^T\mathbf{T}_{i+1})(\tilde{\mathbf{s}}_{>i} + \tilde{\mathbf{a}}_{>i})\right] \\ &= \mathbf{t}_i^T\mathbf{T}_{i+1}^{-1}\mathbf{a}_{>i} - \left[\tilde{\mathbf{s}}_i + (\mathbf{t}_i^T\mathbf{T}_{i+1}^{-1} + \mathbf{r}_i^T)(\mathbf{s}_{>i} + \mathbf{a}_{>i})\right]. \end{aligned} \quad (39)$$

Because $\mathbf{t}_i^T\mathbf{T}_{i+1}^{-1}\mathbf{a}_{>i}$ is also an integer, we arrive at

$$\begin{aligned} \mathbf{a}_i &= -\left[\tilde{\mathbf{s}}_i + \mathbf{t}_i^T\mathbf{T}_{i+1}^{-1}\mathbf{s}_{>i} + \mathbf{r}_i^T(\mathbf{s}_{>i} + \mathbf{a}_{>i})\right] \\ &= -\left[\mathbf{s}_i + \mathbf{r}_i^T(\mathbf{s}_{>i} + \mathbf{a}_{>i})\right] \end{aligned} \quad (40)$$

and, hence, the same equation as in the normal NP algorithm [cf. (14)] arises. It follows that the LLL-aided version of the NP algorithm results in exactly the same solution as the conventional NP algorithm and, therefore, also results in the same performance. $\square$

Corollary 2. *Theorem 1 can also be derived for THP by combining the THP method according to (15) with LR. As the only difference between THP and the NP algorithm is a lower instead of an upper triangular matrix, the same concept of proof can be applied. Hence, also LLL-aided THP with the scaling in (30) cannot outperform THP or NP.*

Remark 2. *Note that Theorem 1 only holds for the optimized $\mathbf{D}$ in (22). For a different $\mathbf{D}$, LR can make a difference. In Section VII, we show that for $\mathbf{D} = \mathbf{I}$, LR significantly improves upon the conventional NP algorithm. However, this is no solution as the rate is clearly degraded in comparison to the optimized matrix in (22).*

B. LLL-aided Rounding Off

Combining the RO algorithm with LR does not lead to the zero vector anymore. While it improves on the conventional algorithm, also the LLL-aided RO algorithm cannot exceed the rate of the conventional NP (or THP) when considering the scaling in (22) [or (30)] which is in accordance with the bounds in [8].

Proposition 3. *The LLL-aided RO algorithm cannot exceed the performance of the conventional NP (or THP) when considering the optimized scaling in (22) [or (30)].*

Proof. We start by using the transmit symbol as in (34) given by

$$\mathbf{x} = \mathbf{H}^+ \mathbf{DT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}}). \quad (41)$$

The RO procedure gives now the simple integer solution $\tilde{\mathbf{a}} = -\lceil \tilde{\mathbf{s}} \rceil$, resulting in the symbol energy

$$\mathbb{E}[\|\tilde{\mathbf{x}}\|^2] = \mathbb{E}[\|\mathbf{H}^+ \mathbf{DT} \text{Mod}(\tilde{\mathbf{s}})\|^2]. \quad (42)$$

With t_{ij}^{-1} being the entries of $\mathbf{T}^{-1}$, we can express $\text{Mod}(\tilde{\mathbf{s}}_i)$ as

$$\text{Mod}(\tilde{\mathbf{s}}_i) = \text{Mod}\left(\mathbf{e}_i^T \mathbf{T}^{-1} \mathbf{s}\right) = \text{Mod}\left(\mathbf{s}_i + \sum_{j>i} t_{ij}^{-1} \mathbf{s}_j\right) \quad (43)$$

since $\mathbf{T}$ is unittriangular. Because $\mathbf{s}_i$ is independent of $\mathbf{s}_j$, we can conclude from Lemma 1 that $\text{Mod}(\tilde{\mathbf{s}}_i) \sim \mathcal{U}(-0.5, 0.5)$. Additionally, as $\tilde{\mathbf{s}}_j$ only depends on $\mathbf{s}_m$, $m \geq j$, and $\mathbf{s}_i$ within $\tilde{\mathbf{s}}_i$ is independent of $\mathbf{s}_j$, $j > i$, it also follows that

$$\text{Mod}(\tilde{\mathbf{s}}_i) | \text{Mod}(\tilde{\mathbf{s}}_j) \sim \mathcal{U}(-0.5, 0.5) \quad \text{for } j > i. \quad (44)$$

The reason is that $\mathbf{s}_i$ within $\tilde{\mathbf{s}}_i$ is independent of the condition, and with Lemma 1, we arrive at (44). This means that $\text{Mod}(\tilde{\mathbf{s}}_i)$ is independent of $\text{Mod}(\tilde{\mathbf{s}}_j)$ for $j > i$. As this holds $\forall j \neq i$,

$$\text{Mod}(\tilde{\mathbf{s}}_i) \text{ is independent of } \text{Mod}(\tilde{\mathbf{s}}_j) \quad \forall j \neq i. \quad (45)$$

Therefore, $\mathbb{E}[\tilde{\mathbf{s}}\tilde{\mathbf{s}}^T] = \mathbf{I}_{\frac{1}{12}}$ and the symbol energy is given by

$$\mathbb{E}[\|\mathbf{H}^+ \mathbf{DT} \text{Mod}(\tilde{\mathbf{s}})\|^2] = \text{tr}\left(\mathbf{T}^T \mathbf{D}(\mathbf{H}\mathbf{H}^T)^{-1} \mathbf{DT}\right) / 12. \quad (46)$$

By using the arithmetic-geometric mean inequality, we obtain

$$\begin{aligned} \mathbb{E}[\|\tilde{\mathbf{x}}\|^2] &\geq 2K^{2K} \sqrt{\det(\mathbf{T}^T \mathbf{D}(\mathbf{H}\mathbf{H}^T)^{-1} \mathbf{DT})} \frac{1}{12} \\ &= \frac{K}{6}^{2K} \sqrt{\det((\mathbf{H}\mathbf{H}^T)^{-1})}^{2K} \sqrt{\prod_{i=1}^{2K} d_{ii}^2}. \end{aligned} \quad (47)$$

Hence, we can upper bound the LLL-RO rate [see (9)] as

$$R_{\text{LLL-RO}}^{\text{HSNR}} \leq \log_2 \det(\bar{\rho} \mathbf{H}_C \mathbf{H}_C^H) - K \log_2 \left(\frac{\pi e}{6}\right). \quad (48)$$

Note that the bound yields the same rate as for the conventional NP (or THP) algorithm. However, this upper bound is, in general, not achievable by the LLL-aided RO method. $\square$

VI. GENERALIZATION TO GREEDY-BASED SCHEMES

In the previous section, we have shown that the classical LR techniques, the LLL-aided RO, and also the LLL-aided NP cannot outperform conventional THP or the conventional NP method. This suggests that LR can be omitted when considering these low-complexity algorithms. However, we will show in this section that this observation can also be extended to more complex algorithms. The important observation is within Theorem 1, where we have shown that a greedy-based optimization leads to the same solution independent of a potential LR. This property arises from the special triangular structure of $\mathbf{R}$ and $\mathbf{T}$, which is generally important beyond the classical Babai NP algorithm. The algorithms for which LR is important or not can be categorized with the following theorem. Note that in this theorem we use the notation of Theorem 1, i.e.,

$$\mathbf{T} = \begin{bmatrix} \mathbf{T}_{<i,a} & \mathbf{T}_{<i,b} \\ \mathbf{0} & \mathbf{T}_i \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} \mathbf{R}_{<i,a} & \mathbf{R}_{<i,b} \\ \mathbf{0} & \mathbf{R}_i \end{bmatrix} \quad (49)$$

where $\mathbf{T}_i$ and $\mathbf{R}_i$ are the upper-triangular blocks starting from the i -th row. These are defined as

$$\mathbf{T}_i = \begin{bmatrix} 1 & \mathbf{t}_i^T \\ \mathbf{0} & \mathbf{T}_{i+1} \end{bmatrix}, \quad \mathbf{R}_i = \begin{bmatrix} 1 & \mathbf{r}_i^T \\ \mathbf{0} & \mathbf{R}_{i+1} \end{bmatrix}. \quad (50)$$

Theorem 2. *Let $\mathbf{R}$ be the unittriangular basis matrix of a lattice $\mathcal{L}$ and $\mathbf{T}$ be the corresponding unimodular integer matrix $\mathbf{T}$ from LR (which is also unittriangular, see Lemma 2). Additionally, let $\min_{\mathbf{a} \in \mathbb{Z}^{2K}} \|\mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})\|_2$ with $\tilde{\mathbf{s}} = \mathbf{T}^{-1}\mathbf{s}$ and $\tilde{\mathbf{a}} = \mathbf{T}^{-1}\mathbf{a}$ be the corresponding lattice problem where $\mathbf{a}$ is the integer-constrained optimization variable. Furthermore, the i -th step of a greedy-based scheme is considered where $\mathbf{a}_{>i} = [a_{i+1}, \dots, a_{2K}]^T$ was already obtained. Then, due to the unittriangular structure of $\mathbf{T}$ and $\mathbf{T}^{-1}$:*

a) $\mathbf{a}_{>i}$ does not depend on $\tilde{a}_i$ and

$$a_i = \tilde{a}_i + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i} \quad (51)$$

b) the rows $i+1, \dots, 2K$ of $\mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})$ are independent of $\tilde{a}_i$ and the i -th one is given by

$$\mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}}) = \tilde{a}_i + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i} + s_i + \mathbf{r}_i^T (\mathbf{s}_{>i} + \mathbf{a}_{>i}) \quad (52)$$

From a) and b), it directly follows that

- $\Rightarrow$ Any neighbor in the lattice $\tilde{a}'_i = \tilde{a}_i + \Delta$ directly corresponds to the neighbor $a'_i = a_i + \Delta$ with $\mathbf{a}_{>i}$ unchanged
- $\Rightarrow$ The optimization of $\tilde{a}_i$ is independent of LR, i.e.,

$$\begin{aligned} \tilde{a}_i^* &= \arg \max_{\tilde{a}_i \in \tilde{\mathbb{A}}_i} |\mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2 \\ \iff a_i^* &= \arg \max_{a_i \in \mathbb{A}_i} |\mathbf{e}_i^T \mathbf{R}(\mathbf{s} + \mathbf{a})|^2 \end{aligned} \quad (53)$$

where $\mathbb{A}_i = \{a_i^{(1)}, \dots, a_i^{(N)}\}$ is the constraint set of a_i and $\tilde{\mathbb{A}}_i = \{a_i^{(1)} - \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i}, \dots, a_i^{(N)} - \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i}\}$ the constraint set of $\tilde{a}_i$. Please note that Theorem 1 is the special case of an unconstrained optimization $\tilde{\mathbb{A}}_i = \mathbb{A}_i = \mathbb{Z}$.

Proof. Point a) follows from the triangular structure of $\mathbf{T}^{-1}$, whereas b) directly follows from the derivations in Theorem 1. Using the same notation, we arrive at [cf. (37)]

$$\begin{aligned} \mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}}) &= \tilde{a}_i + \tilde{s}_i + (\mathbf{t}_i^T + \mathbf{r}_i^T \mathbf{T}_{i+1}) (\tilde{\mathbf{s}}_{>i} + \tilde{\mathbf{a}}_{>i}) \\ &= \tilde{a}_i + \tilde{s}_i + (\mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} + \mathbf{r}_i^T) (\mathbf{s}_{>i} + \mathbf{a}_{>i}) \\ &= \tilde{a}_i + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i} + s_i + \mathbf{r}_i^T (\mathbf{s}_{>i} + \mathbf{a}_{>i}) \end{aligned} \quad (54)$$

since $s_i = \tilde{s}_i + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{s}_{>i}$. The first implication directly follows from (51), whereas the second implication can be seen by [cf. (52)]

$$\begin{aligned} \tilde{a}_i^* &= \arg \max_{\tilde{a}_i \in \tilde{\mathbb{A}}_i} |\mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2 \\ &= \arg \max_{\tilde{a}_i \in \tilde{\mathbb{A}}_i} |\tilde{a}_i + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i} + s_i + \mathbf{r}_i^T (\mathbf{s}_{>i} + \mathbf{a}_{>i})|^2 \\ &= -\mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i} + \arg \max_{\Delta \tilde{a}_i \in \mathbb{A}_i} |\Delta \tilde{a}_i + s_i + \mathbf{r}_i^T (\mathbf{s}_{>i} + \mathbf{a}_{>i})|^2 \\ &= -\mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i} + a_i^* \end{aligned} \quad (55)$$

where we exploited that $\mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i}$ is just an integer offset. $\square$

A. Greedy-Based Schemes

Firstly, we consider the greedy-based schemes which follow the points and implications of Theorem 2.

1) *Nearest Plane (THP)*: The NP algorithm, or the THP algorithm, are probably the most popular greedy schemes. The NP algorithm starts with the integer $\tilde{a}_{2K}$ and then successively optimizes each scalar integer $\tilde{a}_i$ until $\tilde{a}_1$ is reached. According to Theorem 1, this approach results in the same performance regardless of whether LR is applied or not. This can now also be seen by Theorem 2. According to (53), the optimized value in the i -th step $\tilde{a}_i^*$ including LR directly corresponds to the optimal value a_i^* without LR.

2) *K-Best Algorithm*: The K-Best algorithm (see, e.g., [23], [24], [25], [26], [27], [28], [29]) is also a greedy-based scheme and generalizes the concept of the simple NP algorithm. Instead of just successively taking the best integer in each step (as the NP or THP algorithm does), the B best points are considered. These K-Best type of algorithms have already been shown to achieve very good performance close to optimality. However, for optimized performance, it is important that LR is performed (see [27], [28], [29]). The LLL-aided K-Best algorithm starts with the integer $\tilde{a}_{2K}$

and determines the B best candidates. Afterward, for each candidate $\tilde{\mathbf{a}}_{2K}^{(b)}$, $b = 1, \dots, B$, the C best children $\tilde{a}_{2K-1}^{(b,1)}, \dots, \tilde{a}_{2K-1}^{(b,C)}$ are derived in a greedy manner. Then again, the best B candidates are taken of all BC children and the next layer $(2K-1)$ is considered with the candidates $\tilde{\mathbf{a}}_{2K-1}^{(b)}$, $b = 1, \dots, B$. Similarly, the C best children of the B candidates are calculated, and the procedure is continued until the final integer $\tilde{a}_1$ is considered. Lastly, the best candidate is taken and the solution vector $\tilde{\mathbf{a}}^*$ is obtained. This K-Best procedure is a greedy-based scheme, and we can provide the following corollary.

Corollary 3. When the basis matrix $\mathbf{R}$ of a lattice $\mathcal{L}$ is unitriangular (the unimodular integer matrix $\mathbf{T}$ from LR is also unitriangular, see Lemma 2). Then, the LLL-aided K-Best algorithm leads to exactly the same solution as the conventional K-Best algorithm, which happens for lattice precoding with the optimized scaling in (22).

Proof. We assume that we have given the candidates $\tilde{\mathbf{a}}_{i+1}^{(b)}$, $b = 1, \dots, B$ and we are searching for the B best candidates for the i -th integer $\tilde{a}_i$ of a LLL-aided K-Best algorithm. The candidates of the previous layer $\tilde{\mathbf{a}}_{i+1}^{(b)}$, $n = 1, \dots, N$ directly correspond to $\mathbf{a}_{i+1}^{(b)}$, $b = 1, \dots, B$, i.e., the B candidates of a conventional K-Best algorithm without LR. Now, the best C children of each candidate $\tilde{\mathbf{a}}_{i+1}^{(b)}$ have to be derived. According to Theorem 2, optimizing $\|\mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})\|$ for a given $\tilde{\mathbf{a}}_{i+1}^{(b)}$ w.r.t. $\tilde{a}_i$ results in the same integer a_i when maximizing $\|\mathbf{e}_i^T \mathbf{R}(\mathbf{s} + \mathbf{a})\|$ for the given $\mathbf{a}_{i+1}^{(b)}$ without LR. Hence, the optimized integer value $\tilde{a}_i^{(b,1)}$, directly corresponds to $a_i^{(b,1)}$ which is the optimized integer value without LR. The next best children $\tilde{a}_i^{(b,2)}$ are now determined by

$$\tilde{a}_i^{(b,2)} = \arg \max_{\tilde{a}_i \in \mathbb{Z} \setminus \tilde{a}_i^{(b,1)}} |\mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2 \quad (56)$$

and the c -th best, correspondingly, by

$$\tilde{a}_i^{(b,c)} = \arg \max_{\tilde{a}_i \in \mathbb{Z} \setminus \{\tilde{a}_i^{(n,1)}, \dots, \tilde{a}_i^{(b,c-1)}\}} |\mathbf{e}_i^T \mathbf{RT}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2. \quad (57)$$

Based on Theorem 2 the set of the children $\{\tilde{a}_i^{(b,1)}, \dots, \tilde{a}_i^{(b,C)}\}$ directly corresponds to the set $\{a_i^{(b,1)}, \dots, a_i^{(b,C)}\}$ which is obtained with the same procedure without LR. Accordingly, also determining the N best of all NL children results in the same set regardless if LR is applied or not. Hence, we obtain the same set of candidates $\tilde{\mathbf{a}}_i^{(b)}$, $b = 1, \dots, N$ and after successively proceeding with the algorithm, the same solution vector is obtained for the LLL-aided K-Best and conventional K-Best algorithm. Please note that also the computational complexity is exactly the same. $\square$

3) *Sphere Decoder*: The SD is a very popular algorithm, see e.g. [3], [4], [5], [6], [7]. As we are dealing with precoding in the DL, it is also called the sphere encoder (see [2]). While the SD gives the optimal solution of the lattice problem, its exponential complexity renders it impractical. In Algorithm 1, we present the SD according to [3], [4] with the difference that in line 10), the best (in a greedy sense) instead of the first candidate is selected. This is a popular modification in the literature (see [42], [43]) and increases the convergence speed of the SD. While not obvious at first, the SD also belongs to the greedy-based schemes according to Theorem 2, and we will analyze this in the following. As the SD is optimal, it is apparent that with or without LR, the same optimal point is reached. The complexity, on the other hand, is in general reduced when LR is applied (see [22]). However, according to [22], when no column permutations occur during LR, the complexity is,

in general, not reduced. For a unitriangular lattice basis matrix $\mathbf{R}$, it can be shown (see Lemma 2) that no column permutations occur during LR and, hence, no complexity reduction is to be expected. The complexity analysis in [22] is based on an approximation. For our special case of a unitriangular basis matrix, we can extend the analysis in [22] and show that the same sequence of lattice points is visited regardless of whether LR is applied or not. Hence, for the problem we consider, the complexity is actually exactly the same.

Algorithm 1 Sphere Decoder [3], [4]

Require: $r > 0$

Initialize: $i = 2K$

- 1) For fixed $\tilde{\mathbf{a}}_{>i} = [\tilde{a}_{i+1}, \dots, \tilde{a}_{2K}]^T$, find all $\tilde{a}_i \in \mathbb{A}_i$ for which $\|\mathbf{R}_i \mathbf{T}_i(\tilde{\mathbf{s}}_i + \tilde{\mathbf{a}}_i)\|^2 \leq r^2$
 - 2) **if** $\tilde{\mathbb{A}}_i = \emptyset$ **then**
 - 3) $i \leftarrow i + 1$
 - 4) **if** $i = 2K + 1$ **then**
 - 5) Return($\tilde{\mathbf{a}}^*$)
 - 6) **else**
 - 7) GoTo 2)
 - 8) **end if**
 - 9) **end if**
 - 10) Pick $\tilde{a}_i \in \tilde{\mathbb{A}}_i$ with smallest $\|\mathbf{R}_i \mathbf{T}_i(\tilde{\mathbf{s}}_i + \tilde{\mathbf{a}}_i)\|^2$
 - 11) Remove $\tilde{a}_i$ from $\tilde{\mathbb{A}}_i$
 - 12) $i \leftarrow i - 1$
 - 13) **if** $i \geq 1$ **then**
 - 14) GoTo 1)
 - 15) **else**
 - 16) Update $\tilde{\mathbf{a}}^*$
 - 17) Update r
 - 18) $i = 2$
 - 19) GoTo 2)
 - 20) **end if**
-

Corollary 4. When the basis matrix $\mathbf{R}$ of a lattice $\mathcal{L}$ is unitriangular (the unimodular integer matrix $\mathbf{T}$ from LR is also unitriangular, see Lemma 2). Then, the LLL-aided SD algorithm visits exactly the same lattice points in the same order as the conventional SD algorithm. This happens for lattice precoding with the optimized scaling in (22).

Proof. While the SD achieves the optimum performance, it is still based on a greedy procedure in the sense that the search tree is always successively constructed. To see this, we show that the same candidates are generated in line 1) of Algorithm 1 and also selected in line 10). In line 1), the set $\tilde{\mathbb{A}}_i$ is constructed by determining all $\tilde{a}_i$ for which $|\mathbf{e}_i^T \mathbf{R}_i \mathbf{T}_i(\tilde{\mathbf{s}}_i + \tilde{\mathbf{a}}_i)|^2 \leq r^2$ holds (for fixed $\tilde{\mathbf{a}}_{>i} = [\tilde{a}_{i+1}, \dots, \tilde{a}_{2K}]^T$). As $\tilde{a}_i$ corresponds to $a_i = \tilde{a}_i + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i}$ it directly follows that [cf. (52)]

$$|\mathbf{e}_i^T \mathbf{R}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2 = |\tilde{a}_i + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i} + s_i + \mathbf{r}_i^T (s_{>i} + \mathbf{a}_{>i})|^2 \leq r^2$$

$$\iff |a_i + s_i + \mathbf{r}_i^T (s_{>i} + \mathbf{a}_{>i})|^2 \leq r^2 \quad (58)$$

and, hence, the set $\tilde{\mathbb{A}}_i = \{a_i^{(1)} + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i}, \dots, a_i^{(N)} + \mathbf{t}_i^T \mathbf{T}_{i+1}^{-1} \mathbf{a}_{>i}\}$ for which $|\mathbf{e}_i^T \mathbf{R}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2 \leq r^2$ directly corresponds to the set $\mathbb{A}_i = \{a_i^{(1)}, \dots, a_i^{(N)}\}$ of the SD without LR for which $|\mathbf{e}_i^T \mathbf{R}(\mathbf{s} + \mathbf{a})|^2 \leq r^2$. In 1) the $\tilde{a}_i \in \tilde{\mathbb{A}}_i$ is chosen which minimizes the objective function based on a greedy step. This corresponds to $\tilde{a}_i^* = \arg \max_{\tilde{a}_i \in \tilde{\mathbb{A}}_i} |\mathbf{e}_i^T \mathbf{R}(\tilde{\mathbf{s}} + \tilde{\mathbf{a}})|^2$. From Theorem 2, we know that this is equivalent to $a_i^* = \arg \max_{a_i \in \mathbb{A}_i} |\mathbf{e}_i^T \mathbf{R}(\mathbf{s} + \mathbf{a})|^2$. In summary, the same candidates are determined, and the same lattice

points are selected. Hence, the LLL-aided SD and the conventional SD without LR visit the same lattice points. $\square$

B. Counter Examples

Many popular algorithms are based on a greedy procedure and comply with Theorem 2. However, there exist algorithms for which LR does make a difference even when having a unitriangular basis matrix $\mathbf{R}$. These will be examined in the following.

1) *Rounding Off:* We have already analyzed the Rounding Off procedure. The RO algorithm does not belong to the greedy-based methods. Without LR, the solution is the zero-vector according to (13), whereas when considering LR, the optimal solution is, in general, an integer vector different from zero. This also holds for the unitriangular basis matrix, which arises for lattice precoding when considering the optimized scaling in (22). However, we also know from Proposition 3 that the performance cannot exceed the NP or THP algorithm even when LR is applied.

2) *Exhaustive Search:* Exhaustive search is another example of an algorithm not following a greedy-based procedure. For exhaustive search, we restrict each integer value $\tilde{a}_i$ to lie within an interval around some initial point $\tilde{a}_i^{(\text{init})}$ which results in the constraint set $-l_i \leq \tilde{a}_i - \tilde{a}_i^{(\text{init})} \leq u_i$. Afterward, all possible combinations are considered, and the best one is taken. For this method, LR does make a difference in general.

Corollary 5. A LLL-aided exhaustive search algorithm generally leads to a different solution than a conventional exhaustive search, even if the lattice basis matrix is unitriangular, which happens for lattice precoding with the optimized scaling in (22).

Proof. Firstly, we consider a neighboring point which lies within the constraint set by changing the i -th integer as

$$\tilde{a}_i = \tilde{a}_i^{(\text{init})} + \Delta \quad (59)$$

where $\Delta_i \in \{\pm 1, \pm 2, \dots\}$ such that $-l_i \leq \Delta \leq u_i$. According to Theorem 2, this directly corresponds to the neighboring point

$$a_i = a_i^{(\text{init})} + \Delta \quad (60)$$

of $a_i^{(\text{init})}$ without LR with $\mathbf{a}_{>i}$ unchanged. However, the integer values $\mathbf{a}_{<i}$ are implicitly also changed by adding Δ to $a_i^{(\text{init})}$, which results from the exhaustive search not belonging to the greedy-based procedures. Actually, the integers a_j with $j < i$ are given as

$$\begin{aligned} a_j &= \tilde{a}_j^{(\text{init})} + \mathbf{t}_j^T \tilde{\mathbf{a}}_{>j} \\ &= \tilde{a}_j^{(\text{init})} + \mathbf{t}_j^T (\tilde{\mathbf{a}}_{>j}^{(\text{init})} + \mathbf{e}_{i-j} \Delta) \\ &= a_j^{(\text{init})} + t_{ji} \Delta. \end{aligned} \quad (61)$$

As t_{ji} is an arbitrary integer and can have any value, $t_{ji} \Delta$ is, in general, non-zero. Even more, $a_j^{(\text{init})} + t_{ji} \Delta$ might actually lie outside the constraint set $-l_i \leq a_i - a_i^{(\text{init})} \leq u_i$ of the conventional exhaustive search without LR. Therefore, in comparison to greedy-based methods, the neighbors are not preserved and the constraint set $-l_i \leq \tilde{a}_i - \tilde{a}_i^{(\text{init})} \leq u_i$ does not correspond to the set $-l_i \leq a_i - a_i^{(\text{init})} \leq u_i$. $\square$

3) *Fixed-Complexity Sphere Decoder:* Due to the exponential complexity, exhaustive search is typically not employed for lattice problems. However, there are also more practical methods where LR still has an effect. One of them is the FSD [30], [31], [32], [33], which showed very good performance for the detection problem. This approach has also been applied for precoding (see [34], [35], [36]), where it is also called the Fixed-Complexity Sphere Encoder. For the

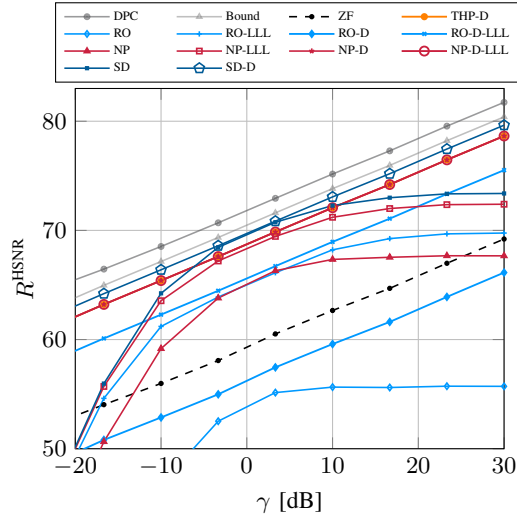


Fig. 1. $N = 6$, $P_{Tx} = 40$ dB

FSD, the number of nodes in each layer is fixed, where the specific number is a design choice. Typically, the integers $N+1, \dots, 2K$ are determined by exhaustive search and then, for all these candidates, the NP algorithm is applied for the remaining layers (see [30], [34]). We observed that the optimal lattice point is typically very close to the LLL-RO or the (LLL)-NP solution vector. Hence, we restrict the search space of each element close to an initial solution (like the NP solution), i.e., $-\Delta \leq a_i - a_i^{(init)} \leq \Delta$. By choosing now a smaller number for Δ , the number of integers for exhaustive search that can be computationally afforded increases, and we have observed a clearly improved performance in our scenarios. As the layers $N+1, \dots, 2K$ are determined by exhaustive search, the FSD does not belong to the greedy-based methods, which we summarize in the following corollary.

Corollary 6. *A LLL-aided FSD generally leads to a different solution than a conventional FSD search, even if the lattice basis matrix is unitriangular, which happens for lattice precoding with the optimized scaling in (22).*

Proof. As the first layers $N+1, \dots, 2K$ are determined by exhaustive search, the FSD generally leads to a different solution when LR is applied. $\square$

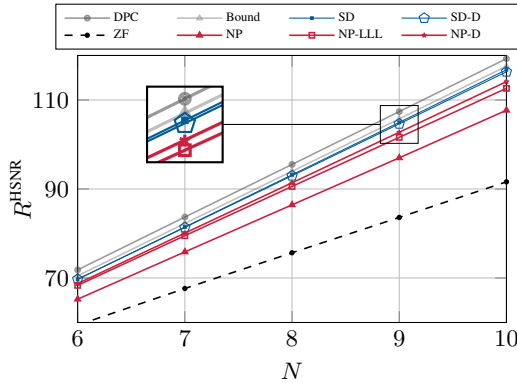


Fig. 2. $\gamma = 0$ dB, $P_{dB} = 40$ dB

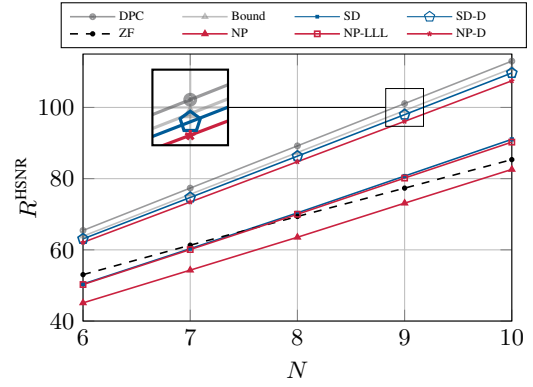


Fig. 3. $\gamma = -20$ dB, $P_{dB} = 40$ dB

VII. NUMERICAL RESULTS

We now support the theoretical results of the last sections with simulations. We consider a quadratic system where $\mathbf{H}_C \in \mathbb{C}^{K \times K}$ has i.i.d. $\mathcal{N}_C(0, 1)$ entries. To analyze for worse conditioning, we impose a path loss γ to user 1. Prior to applying the algorithms, a coding order optimization w.r.t. the mean squared error (MSE) is performed (see [13]), which is similar to a V-BLAST ordering. This is especially important for the moderate and low SNR regions.

A. Classical Algorithms

For the algorithms, we use linear ZF and DPC, both with Gaussian symbols, as a reference. All other methods have uniformly distributed symbols and, hence, experience a shaping loss. As an upper bound (Bound), we use expression (11) from [18] based on the results in [21]. For the algorithms, we start with an identity scaling $\mathbf{D}$ and consider the RO and NP algorithms, their LLL-aided versions RO-LLL and NP-LLL, as well as the SD algorithm. Additionally, we analyze the RO and NP methods with their optimal scaling matrix $\mathbf{D}$ from (16) and (22) denoted as RO-D and NP-D, respectively. Furthermore, we use the optimized scaling in (22) with the LLL-aided RO and NP algorithms together with the SD algorithm stated as RO-D-LLL, NP-D-LLL, and SD-D, respectively. Lastly, we consider THP-D with the optimal scaling in (30). For the LLL algorithm, we use the implementation of [15].

In Fig. 1, the high-SNR asymptotes of all algorithms are compared with each other for the dimension $N = K = 6$. To see the importance of the matrix $\mathbf{D}$, the pathloss γ of user 1 is modified. We can

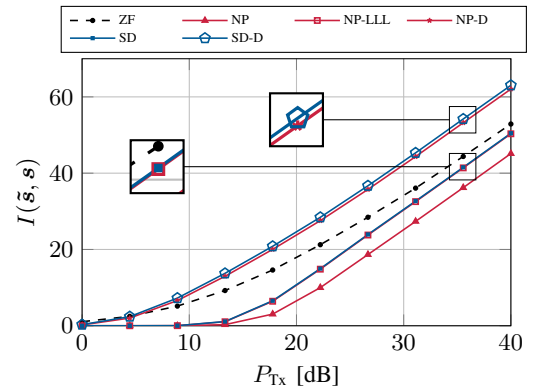


Fig. 4. $N = 6$, $\gamma = -20$ dB

observe that all algorithms including the scaling in (22) [or (30) for THP-D] are robust w.r.t. a change of γ . On the other hand, the algorithms relying on the identity scaling, experience a significant degradation in performance when γ is modified. Only for around $\gamma = 0$ dB, which corresponds to a well-conditioned channel, the identity scaling $\mathbf{D} = \mathbf{I}$ is valid for this scenario. For smaller values, the performance decreases rapidly, whereas for higher values, the rate saturates even though the channel gain increases. The optimality of the scaling in (22) has been shown in Proposition 2 in the case of the NP algorithm. However, we can see that this is, in general, a good choice, as also the SD algorithm benefits from it. The SD-D algorithm clearly outperforms the SD algorithm, with the difference being especially pronounced if the channel is not well conditioned. Beyond the advantages of the scaling in (22) [or (30)], Fig. 1 additionally allows to verify Theorem 1 and Corollary 2 by recognizing that THP-D, NP-D, and NP-D-LLL have the same performance. As stated in Remark 2, this only holds for the algorithms relying on the scaling in (22) [or (30)]. For the algorithms with identity scaling, we can see that LR has a strong impact, and NP-LLL leads to a significant performance increase in comparison to NP. However, as addressed in Remark 2, the performance of NP-LLL (and therefore also NP) is worse in comparison to NP-D-(LLL). Furthermore, we can see that all RO algorithms lead to a worse performance than their NP counterpart. LR leads to a performance increase for all RO versions with RO-D-LLL, i.e., RO with the scaling in (22) together with LR, achieving the best performance. Nevertheless, verifying Proposition 3, RO-D-LLL cannot achieve a better performance than NP-D-(LLL) (or THP-D).

As NP-D-LLL, NP-D, and THP-D lead to exactly the same performance, we will only consider NP-D in the following figures. Additionally, as the RO algorithms lead to a worse performance in comparison to their NP counterparts, the RO methods are also omitted in the following.

We will now further analyze the importance of the scaling matrix $\mathbf{D}$ and the influence of the channel condition. According to Fig. 1, the channel condition has a huge impact on the choice of $\mathbf{D}$. We have observed that the identity scaling leads to a significant degradation in the performance for all algorithms. Only for well-conditioned channels, the identity matrix seems to be a good choice. This is further illustrated in Figs. 2 and 3, where we can see the high-SNR asymptotes w.r.t. the number of base station antennas (equal to the number of users). For the well-conditioned channels in Fig. 2 ($\gamma = 0$ dB), the scaling matrix $\mathbf{D}$ has only a slight impact on the performance. However, when decreasing the channel gain of user 1 in Fig. 3 ($\gamma = -20$ dB), this is different, and the impact of optimizing the scaling matrix $\mathbf{D}$ is very pronounced. This holds for all methods, including the SD algorithm. Actually, even the low-complexity method NP-D with the optimal scaling in (22) clearly outperforms the SD with the identity scaling $\mathbf{D} = \mathbf{I}$.

The observations above are also valid when considering the actual rate expressions according to (8) instead of the high-SNR asymptotes. In Fig. 4, we can see that all curves converge to their asymptotes for a high transmit power. Additionally, it is important to note that the identity scaling is also problematic for moderate SNR values, leading to a later onset of the methods w.r.t. the power.

B. Advanced Algorithms

Having analyzed the classical LR techniques like the RO or NP algorithm, we are now considering more advanced techniques. In the following simulations, we are focusing on the K-Best and the FSD, which we discussed in Section VI. For the K-Best algorithm, we take the B candidates in each step, where, if not stated otherwise, we set

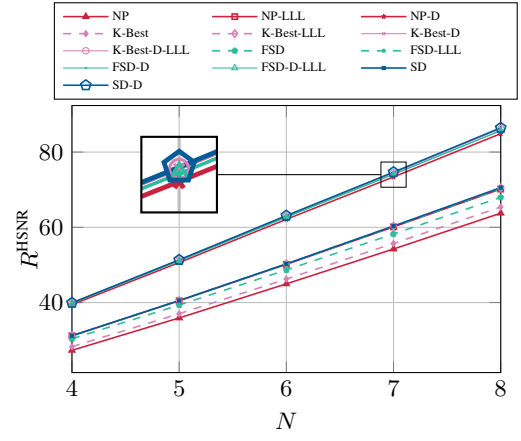


Fig. 5. $\gamma = -20$ dB, $\kappa = 0$, $P_{Tx} = 40$ dB, $B = D = 3$

$B = 3$. Additionally, each candidate is expanded with 3 children in each step. Please see Section VI for details. For the FSD, we use D dimensions, i.e., $a_{2K-D+1}, \dots, a_{2K}$ for exhaustive search whereas the remaining dimensions, i.e., $a_{2K-D}, \dots, a_1$, are given by the NP algorithm for each of the possible candidates. The exhaustive search is initialized with the NP-(D)-LLL solution, i.e., the constraint set $-\Delta \leq a - a_i^{NP-(D)-LLL} \leq \Delta$ is considered for $i = 2K-D+1, \dots, 2K$. We choose $\Delta = 1$ as we have observed that the solution is typically close to the NP-(D)-LLL solution. This small value of Δ allows a higher number of D , i.e., the dimensions with exhaustive search. We have experienced a better performance with a small Δ and higher D . Please see Section VI for details.

The K-Best as well as the FSD are analyzed with and without LR and, additionally, with and without the scaling matrix in (22). This results in the algorithms, K-Best, K-Best-LLL, K-Best-D, K-Best-D-LLL for the K-Best methods and in the algorithms FSD, FSD-LLL, FSD-D, FSD-D-LLL for the FSD methods. To allow a better comparison, we keep the SD-(D) as well as the NP-(D)-(LLL) methods. Please note again that the NP-D-LLL, NP-D, and the THP-D have the same performance and, hence, only NP-D is given in the following plots.

In Fig. 5, we analyze the same setup as in Fig. 3, where one user has an additional pathloss of 20 dB. We can again observe that the algorithms with the scaling in (22) achieve a much better performance than the algorithms with an identity scaling. This time, the focus is on

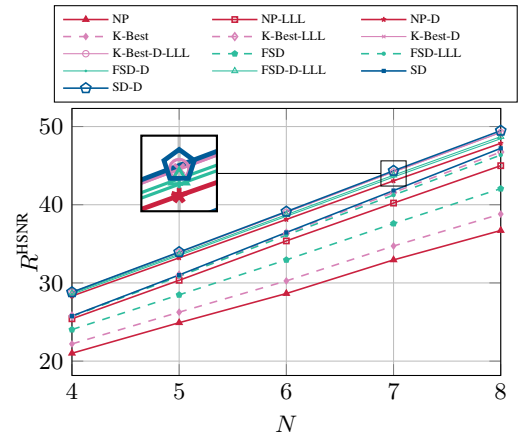


Fig. 6. $\gamma = 0$ dB, $\kappa = 20$ dB, $P_{Tx} = 40$ dB, $B = D = 3$

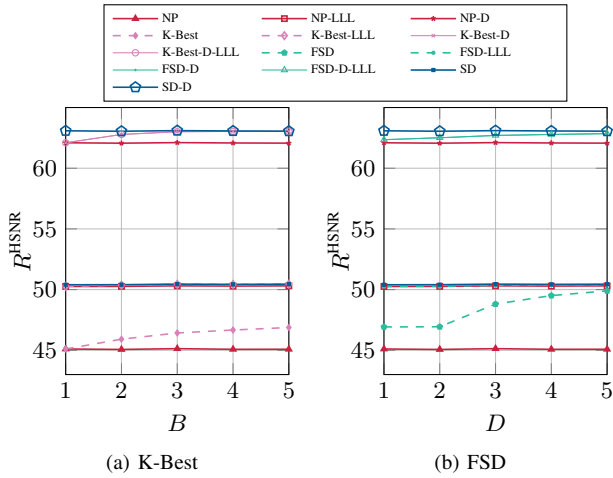


Fig. 7. $N = 6$, $\gamma = -20$ dB, $\kappa = 0$ dB, $P_{Tx} = 40$ dB

the more advanced techniques, i.e., the K-Best and the FSD. In case of an identity scaling, the conventional K-Best and the FSD are both worse than their LLL-aided versions, K-Best-LLL and FSD-LLL. The LLL-aided versions have a similar performance, almost overlapping with the optimal SD. Without LR, the performance is worse for both algorithms, with the FSD achieving a better performance than the K-Best method. For the methods with the scaling in (22), the situation is different. Here, there is no difference between the conventional and the LLL-aided versions. For the K-Best algorithms, this has been shown in Section VI, whereas for the FSD, this can only be observed for this specific scenario. Additionally, the K-Best algorithms K-Best-D and K-Best-D-LLL lead to a slightly better performance in comparison to FSD-D and FSD-D-LLL.

We are now considering a different scenario. In the previous simulation, the condition of the channel matrix was reduced by adding an extra path loss of 20 dB for one user. Now, the condition is reduced differently. Instead of adding an extra path loss, the correlation of the users is increased. For this, we are using a Rician model. As the specific scenario is not important for our purposes, we select the all-ones vectors for simplicity and use the channel model

$$\mathbf{H}_C = \sqrt{\frac{\kappa}{1+\kappa}} \mathbf{1}\mathbf{1}^T + \sqrt{\frac{1}{1+\kappa}} \mathbf{H}_C^{\text{Ray}} \quad (62)$$

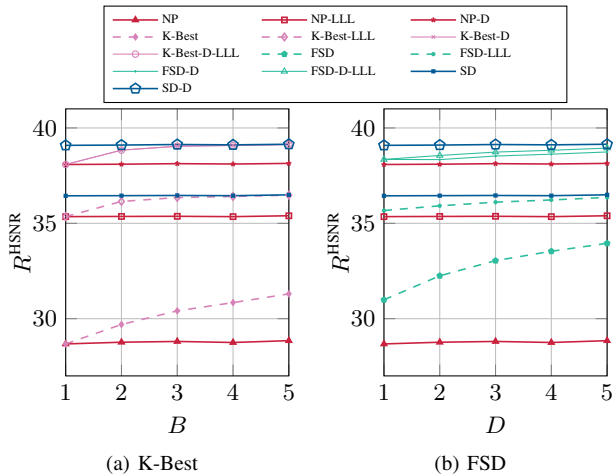


Fig. 8. $N = 6$, $\gamma = 0$ dB, $\kappa = 20$ dB, $P_{Tx} = 40$ dB

where $\mathbf{H}_C^{\text{Ray}} \in \mathbb{C}^{K \times K}$ has i.i.d. $\mathcal{N}_C(0, 1)$ entries and κ is the Rician factor, controlling the condition of the channel matrix.

Fig. 6 shows the performance of the different methods for a Rician factor of $\kappa = 20$ dB. We observe again that the methods with the identity scaling perform worse in comparison to the ones based on (22). However, the more advanced LLL-aided methods K-Best-LLL, FSD-LLL, and especially the SD algorithm improve for an increasing dimension of the channel matrix. For the simple NP-LLL, this improvement cannot be observed. As in the scenario before, LR improves the algorithms considerably in the case of the identity scaling. Interestingly, the K-Best leads again to a worse performance than FSD. However, the LLL-aided version K-Best-LLL achieves a higher rate than FSD-LLL. When focusing on the algorithms based on the scaling (22), we see again that the K-Best-D achieves the same rate as K-Best-D-LLL according to Section VI and, additionally, leads to a better performance than the FSD methods FSD-D and FSD-D-LLL. In comparison to Fig. 5, a slight difference appears between FSD-D-LLL and FSD-D.

We analyze now the performance of the K-Best and FSD method w.r.t. their complexity parameters B and D for both scenarios above and a dimension of $N = 6$. Starting with the i.i.d. Rayleigh scenario where one user has an additional pathloss of 20 dB, we can see the dependence on the parameters B and D for K-Best and FSD in Fig. 7. We can see that the K-Best improves a lot faster than the FSD. It has to be noted that the computational complexity scales polynomially for K-Best with B but exponentially for the FSD with D . Only the K-Best with the identity scaling and without LR has a poor performance, significantly worse than the FSD method (also with identity scaling and without LR).

The same behaviour can be seen in Fig. 8 where all users have the same pathloss, but a Rician model with Rician factor $\kappa = 20$ dB is considered. One difference to Fig. 7 is the difference of FSD-D and FSD-D-LLL, which is not seen in Fig. 7 where the channels are uncorrelated.

VIII. CONCLUSION

In this article, we have demonstrated that there are clear differences for VP in the DL when considering the mutual information in comparison to the SER and BER as the metric. We have seen that a diagonal scaling in front of the pseudoinverse when constructing the precoder at the BS is very important, leading to very different rates. With the wrong scaling, even very simple schemes like the THP can outperform the SD. This is especially important for ill-conditioned channels. Additionally, we have seen that when using an optimized scaling, popular LR techniques like the Babai RO and the Babai NP algorithms cannot outperform simple schemes like THP or the NP algorithm without LR. This can also be generalized to more advanced algorithms like the K-Best algorithm (or the SD), where LR also does not make any difference. The K-Best algorithm worked very well for our scenarios and has performed better than the FSD, which we also considered in the simulations. Future work will analyze more complex precoders than the ZF, like, e.g., the MMSE precoder in a moderate SNR regime. Additionally, low-complexity algorithms specifically designed for the case of the optimized scaling (unitriangular basis matrix) will be examined.

REFERENCES

- [1] H. Weingarten, Y. Steinberg, and S. Shamai, "The Capacity Region of the Gaussian Multiple-Input Multiple-Output Broadcast Channel," *IEEE Transactions on Information Theory*, vol. 52, no. 9, pp. 3936–3964, 2006.

- [2] B. Hochwald, C. Peel, and A. Swindlehurst, "A vector-perturbation technique for near-capacity multi-antenna multiuser communication-part ii: perturbation," *IEEE Transactions on Communications*, vol. 53, no. 3, pp. 537–544, 2005.
- [3] E. Viterbo and J. Boutros, "A universal lattice code decoder for fading channels," *IEEE Transactions on Information Theory*, vol. 45, no. 5, pp. 1639–1642, 1999.
- [4] U. Fincke and M. Pohst, "Improved Methods for Calculating Vectors of Short Length in a Lattice, Including a Complexity Analysis," *Mathematics of Computation*, vol. 44, no. 170, pp. 463–471, 1985. [Online]. Available: <http://www.jstor.org/stable/2007966>
- [5] M. Damen, H. El Gamal, and G. Caire, "On maximum-likelihood detection and the search for the closest lattice point," *IEEE Transactions on Information Theory*, vol. 49, no. 10, pp. 2389–2402, 2003.
- [6] A. Burg, M. Borgmann, M. Wenk, M. Zellweger, W. Fichtner, and H. Bölcskei, "VLSI implementation of MIMO detection using the sphere decoding algorithm," *IEEE Journal of Solid-State Circuits*, vol. 40, no. 7, pp. 1566–1577, 2005.
- [7] B. Hassibi and H. Vikalo, "On the sphere-decoding algorithm I. Expected complexity," *IEEE Transactions on Signal Processing*, vol. 53, no. 8, pp. 2806–2818, 2005.
- [8] L. Babai, "On Lovász' lattice reduction and the nearest lattice point problem," *Combinatorica*, vol. 6, no. 1, pp. 1–13, March 1986. [Online]. Available: <https://doi.org/10.1007/BF02579403>
- [9] A. K. Lenstra, H. W. Lenstra, and L. Lovász, "Factoring polynomials with rational coefficients," *Mathematische Annalen*, vol. 261, no. 4, pp. 515–534, December 1982. [Online]. Available: <https://doi.org/10.1007/BF01457454>
- [10] M. Tomlinson, "New automatic equaliser employing modulo arithmetic," *Electronics Letters*, vol. 7, pp. 138–139(1), March 1971.
- [11] H. Harashima and H. Miyakawa, "Matched-Transmission Technique for Channels With Intersymbol Interference," *IEEE Transactions on Communications*, vol. 20, no. 4, pp. 774–780, 1972.
- [12] R. Wesel and J. Cioffi, "Achievable rates for Tomlinson-Harashima precoding," *IEEE Transactions on Information Theory*, vol. 44, no. 2, pp. 824–831, 1998.
- [13] M. Joham, D. A. Schmidt, J. Brehmer, and W. Utschick, "Finite-Length MMSE Tomlinson-Harashima Precoding for Frequency Selective Vector Channels," *IEEE Transactions on Signal Processing*, vol. 55, no. 6, pp. 3073–3088, 2007.
- [14] J. Nossek, M. Joham, and W. Utschick, "Transmit processing in MIMO wireless systems," in *Proceedings of the IEEE 6th Circuits and Systems Symposium on Emerging Technologies: Frontiers of Mobile and Wireless Communication (IEEE Cat. No.04EX710)*, vol. 1, 2004, pp. I–18.
- [15] D. Wübben, D. Seethaler, J. Jaldén, and G. Matz, "Lattice Reduction," *IEEE Signal Processing Magazine*, vol. 28, no. 3, pp. 70–91, 2011.
- [16] C. Windpassinger, R. Fischer, and J. Huber, "Lattice-reduction-aided broadcast precoding," *IEEE Transactions on Communications*, vol. 52, no. 12, pp. 2057–2060, 2004.
- [17] M. Taherzadeh, A. Mobasher, and A. K. Khandani, "Communication Over MIMO Broadcast Channels Using Lattice-Basis Reduction," *IEEE Transactions on Information Theory*, vol. 53, no. 12, pp. 4567–4582, 2007.
- [18] A. Razi, D. J. Ryan, I. B. Collings, and J. Yuan, "Sum rates, rate allocation, and user scheduling for multi-user MIMO vector perturbation precoding," *IEEE Transactions on Wireless Communications*, vol. 9, no. 1, pp. 356–365, 2010.
- [19] A. Razi, D. J. Ryan, J. Yuan, and I. B. Collings, "Sum rates for multi-user MIMO vector perturbation precoding with regularization," *Physical Communication*, vol. 13, pp. 187–196, 2014. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1874490714000731>
- [20] Y. Avner, B. M. Zaidel, and S. Shamai Shitz, "On Vector Perturbation Precoding for the MIMO Gaussian Broadcast Channel," *IEEE Transactions on Information Theory*, vol. 61, no. 11, pp. 5999–6027, 2015.
- [21] D. J. Ryan, I. B. Collings, I. V. L. Clarkson, and R. W. Heath Jr., "A lattice-theoretic analysis of vector perturbation for multi-user MIMO systems," in *2008 IEEE International Conference on Communications*, 2008, pp. 3340–3344.
- [22] X.-W. Chang, J. Wen, and X. Xie, "Effects of the LLL Reduction on the Success Probability of the Babai Point and on the Complexity of Sphere Decoding," *IEEE Transactions on Information Theory*, vol. 59, no. 8, pp. 4915–4926, 2013.
- [23] K. Wong, C. Tsui, R. Cheng, and W. Mow, "A VLSI architecture of a K-best lattice decoding algorithm for MIMO channels," in *2002 IEEE International Symposium on Circuits and Systems (ISCAS)*, vol. 3, 2002, pp. III–III.
- [24] Z. Guo and P. Nilsson, "Algorithm and implementation of the K-best sphere decoding for MIMO detection," *IEEE Journal on Selected Areas in Communications*, vol. 24, no. 3, pp. 491–503, 2006.
- [25] Q. Li and Z. Wang, "Improved k-best sphere decoding algorithms for MIMO systems," in *2006 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2006, pp. 4 pp.–1162.
- [26] M. Joham, L. G. Barbero, T. Lang, W. Utschick, J. Thompson, and T. Ratnarajah, "FPGA implementation of MMSE metric based efficient near-ML detection," in *2008 International ITG Workshop on Smart Antennas*, 2008, pp. 139–146.
- [27] X.-F. Qi and K. Holt, "A Lattice-Reduction-Aided Soft Demapper for High-Rate Coded MIMO-OFDM Systems," *IEEE Signal Processing Letters*, vol. 14, no. 5, pp. 305–308, 2007.
- [28] M. Shabany and P. G. Gulak, "The application of lattice-reduction to the k-best algorithm for near-optimal MIMO detection," in *2008 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2008, pp. 316–319.
- [29] Q. Zhou and X. Ma, "An improved LR-aided k-best algorithm for MIMO detection," in *2012 International Conference on Wireless Communications and Signal Processing (WCSP)*, 2012, pp. 1–5.
- [30] L. G. Barbero and J. S. Thompson, "Fixing the Complexity of the Sphere Decoder for MIMO Detection," *IEEE Transactions on Wireless Communications*, vol. 7, no. 6, pp. 2131–2142, 2008.
- [31] J. Jaldén, L. G. Barbero, B. Ottersten, and J. S. Thompson, "The Error Probability of the Fixed-Complexity Sphere Decoder," *IEEE Transactions on Signal Processing*, vol. 57, no. 7, pp. 2711–2720, 2009.
- [32] L. G. Barbero and J. S. Thompson, "Extending a Fixed-Complexity Sphere Decoder to Obtain Likelihood Information for Turbo-MIMO Systems," *IEEE Transactions on Vehicular Technology*, vol. 57, no. 5, pp. 2804–2814, 2008.
- [33] C. Zheng, X. Chu, J. McAllister, and R. Woods, "Real-Valued Fixed-Complexity Sphere Decoder for High Dimensional QAM-MIMO Systems," *IEEE Transactions on Signal Processing*, vol. 59, no. 9, pp. 4493–4499, 2011.
- [34] M. Mohaisen and K. Chang, "Fixed-complexity sphere encoder for multi-user MIMO systems," *Journal of Communications and Networks*, vol. 13, no. 1, pp. 63–69, 2011.
- [35] M. Barrenechea, L. Barbero, M. Mendicute, and J. Thompson, "Design and hardware implementation of a low-complexity multiuser vector precoder," in *2010 Conference on Design and Architectures for Signal and Image Processing (DASIP)*, 2010, pp. 160–167.
- [36] M. Barrenechea, M. Mendicute, J. Del Ser, and J. Thompson, "Wiener filter-based fixed-complexity vector precoding for the MIMO downlink channel," in *2009 IEEE 10th Workshop on Signal Processing Advances in Wireless Communications*, 2009, pp. 216–220.
- [37] G. Forney, M. Trott, and S.-Y. Chung, "Sphere-bound-achieving coset codes and multilevel coset codes," *IEEE Transactions on Information Theory*, vol. 46, no. 3, pp. 820–850, 2000.
- [38] M. Barrenechea, M. Joham, M. Mendicute, and W. Utschick, "Analysis of vector precoding at high SNR: Rate bounds and ergodic results," in *2010 IEEE Global Telecommunications Conference GLOBECOM 2010*, 2010, pp. 1–5.
- [39] K. Zu, R. C. de Lamare, and M. Haardt, "Multi-Branch Tomlinson-Harashima Precoding Design for MU-MIMO Systems: Theory and Algorithms," *IEEE Transactions on Communications*, vol. 62, no. 3, pp. 939–951, 2014.
- [40] D. Semmler, M. Joham, and W. Utschick, "Nonlinear precoding in the RIS-aided MIMO broadcast channel," 2024. [Online]. Available: <https://arxiv.org/abs/2409.02464>
- [41] D. Wübben, R. Bohnke, V. Kuhn, and K.-D. Kammeyer, "MMSE-based lattice-reduction for near-ML detection of MIMO systems," in *ITG Workshop on Smart Antennas (IEEE Cat. No.04EX802)*, 2004, pp. 106–113.
- [42] C. P. Schnorr and M. Euchner, "Lattice Basis Reduction: Improved Practical Algorithms and Solving Subset Sum Problems," *Mathematical Programming*, vol. 66, no. 1, pp. 181–199, 1994. [Online]. Available: <https://doi.org/10.1007/BF01581144>
- [43] E. Agrell, T. Eriksson, A. Vardy, and K. Zeger, "Closest point search in lattices," *IEEE Transactions on Information Theory*, vol. 48, no. 8, pp. 2201–2214, 2002.